

E-Badge 4.1.0

Features

- Dual power roles on USB-CDC-DRD port
- VIN input from 2.8V to 16V
- Undervoltage and overvoltage protection.
- Overcurrent protection.
- 10W collective power budget from 5V and 3.3V rail. ⁱ

Changes

- Adjusted Joystick footprint to match 3d model.
- Moved CN1 connector to not interfere with joystick.
- Adjusted resistors for RGB LED to lower brightness.
- Adjusted resistors for Red LED to increase brightness.
- Changed out previous 16mA per color channel Addressable LEDs for 5mA per color channel LEDs.
- Changed potentiometer to 500ohm 1 turn potentiometer. (This should make it much more responsive). ⁱⁱ
- Selected power path components to allow for increased current through the board. (More than 2A of current draw).
- Added automatic USB power role switching. The board can be powered from any source and source up 500mA of current to the USB-CDC-DRD port.
- Added VIN screw terminal input with respective components to allow for an external input voltage supply of 2.8V to 16V.
- Swapped 3.3V LDO for buck converter.
- Changed VBAT on E-badge to 5V. ⁱⁱⁱ

ⁱ The 5V rail can source 10W (5V at 2A). The 3.3V rail can source 6.6W (3.3V at 2A) but draws power from the 5V rail pulling from the 10W budget.

ⁱⁱ In my tests the buzzer seemed sufficiently load. But I was driving it with stm32 PWM rather than the ESP. Also, the buzzer is designed to resonate a 4khz so any other frequency will be significantly attenuated.

ⁱⁱⁱ Changing this to 5V meets the E-badge specifications and allows for simplified routing and power management.